

Mathletes Formulas

Sheldon Mathletes

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1. Polynomials

1.1. Linear Equations

For polynomials of degree one, the variable m is used to refer to the slope of the lines drawn by their graphs. Parallel lines have equivalent slopes, while perpendicular lines have slopes that are reciprocals of one another ($m \perp \frac{1}{m}$).

Slope-Intercept

$$y = mx + b$$

Point-Slope

$$y - y_1 = m(x - x_1)$$

Slope

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-Point

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

1.2. Quadratics

Quadratic Equations

Standard Form $ax^2 + bx + c$

Let $D = b^2 - 4ac$,

Vertex Form $a(x - h)^2 + k$

1. $D > 0 \implies 2$ real solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. $D = 0 \implies 1$ real solution

3. $D < 0 \implies$ no real solutions

Minimum and Maximum Values

Consider the function $f(x) = ax^2 + bx + c$ with vertex $(-\frac{b}{2}a, f(-\frac{b}{2}a))$

1. $a > 0 \implies f$ has a *minimum* at $x = -\frac{b}{2}a$. The minimum value is $f(-\frac{b}{2}a)$.

2. $a < 0 \implies f$ has a *maximum* at $x = -\frac{b}{2}a$. The maximum value is $f(-\frac{b}{2}a)$.

1.3. General Polynomials

Let n be a non-negative integer and let $a_n, a_{n-1}, \dots, a_2, a_1, a_0$ be real numbers such that $a_n \neq 0$. The function:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

is a polynomial function of x with degree n .

Linear Factorization Theorem

If $f(x)$ is a polynomial of degree n , where $n > 0$, then $f(x)$ has precisely n linear factors

$$f(x) = a_n(x - c_1)(x - c_2)\dots(x - c_n)$$

where $c_1, c_2, \dots, c_n$ are complex numbers.

Rational Zero Test

If the polynomial has *integer* coefficients, then every rational zero of f has the form $\frac{p}{q}$, where p and q have no common factors other than 1, and

p = a factor of the constant term a_0

q = a factor of the leading coefficient a_n

Conjugate Pairs

Let f be a polynomial function with *real coefficients*. If $a + bi$, where $b \neq 0$, is zero of f , the complex conjugate $a - bi$ is also a zero of f .

For polynomials with *integer coefficients*, $a + \sqrt{b}$ being a zero implies $a - \sqrt{b}$ is also a zero.

Roots

Let f be the polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

$$\text{The sum of all roots} = -\frac{a_n}{a_{n-1}}$$

$$\text{The product of all roots} = (-1)^n \frac{a_0}{a_n}$$

1.4. Rational Functions

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0}$$

where $N(x)$ and $D(x)$ have no common factors.

1. The graph of f has *vertical asymptotes* at the zeros of $D(x)$.
2. The graph of f has at most one *horizontal asymptote* determined by comparing the degrees of $N(x)$ and $D(x)$.
 - a. $n < 0 \implies$ the line $y = 0$ is the horizontal asymptote.
 - b. $n = 0 \implies$ the line $y = \frac{a_n}{b_m}$ is the horizontal asymptote.
 - c. $n > 0 \implies$ there is no horizontal asymptote.

Suppose $N(x)$ and $D(x)$ had some common factor $(x - c)$. The graph of f would then have a hole at $(c, f(c))$

2. Trigonometry

2.1. Functions

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos x = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan x = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin x}{\cos x}$$

$$\csc x = \frac{1}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\cot x = \frac{1}{\tan x}$$

Function	Domain	Range
$\sin^{-1}$	$[-1, 1]$	$[-90^\circ, 90^\circ]$
$\cos^{-1}$	$[-1, 1]$	$[0^\circ, 180^\circ]$
$\tan^{-1}$	$\mathbb{R}$	$[-90^\circ, 90^\circ]$
$\cot^{-1}$	$\mathbb{R}$	$[0^\circ, 180^\circ]$

	0°	30°	45°	60°	90°	180°	270°	360°
sin	0	1	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0	1
cot	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef	0	undef	0

2.2. Identities

Radians and Degrees

$$n^\circ = \frac{\pi}{180}n \text{ rad}$$

$$m \text{ rad} = \left(\frac{180}{\pi}m\right)^\circ$$

Co-Function

$$\sin x = \cos(90 - x)$$

$$\sec x = \csc(90 - x)$$

$$\tan x = \cot(90 - x)$$

Pythagorean

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Even/Odd

$$\sin(-x) = -\sin x \quad \csc(-x) = -\csc x$$

$$\cos(-x) = \cos x \quad \sec(-x) = \sec x$$

$$\tan(-x) = -\tan x \quad \cot(-x) = -\cot x$$

Sum-to-Difference

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \pm \sin x \sin y$$

$$\tan(x \pm y) = \frac{\sin(x \pm y)}{\cos(x \pm y)}$$

Half-Angle

$$\sin\left(\frac{x}{2}\right) = \pm \sqrt{\frac{1 - \cos x}{2}} *$$

$$\cos\left(\frac{x}{2}\right) = \pm \sqrt{\frac{1 + \cos x}{2}} *$$

$$\tan\left(\frac{x}{2}\right) = \pm \frac{\sin(x \div 2)}{\cos(x \div 2)} *$$

Double-Angle

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan 2x = \frac{\sin 2x}{\cos 2x}$$

* where you consider the angle's location to determine the sign

Sum-to-Product

$$\sin x + \sin y = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$$

$$\sin x - \sin y = 2 \sin\left(\frac{x-y}{2}\right) \cos\left(\frac{x+y}{2}\right)$$

$$\cos x + \cos y = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$$

$$\cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

$$\tan x \pm \tan y = \frac{\sin(x \pm y)}{\cos x \cos y}$$

where $\frac{x}{90^\circ} \bmod 2 \neq 1$

$$\cot x \pm \cot y = \frac{\sin(y \pm x)}{\sin x \sin y}$$

where $\frac{x}{90^\circ} \bmod 2 \neq 2$

Product-to-Sum

$$\sin x \sin y = \frac{\cos(x - y) - \cos(x + y)}{2}$$

$$\cos x \cos y = \frac{\cos(x + y) + \cos(x - y)}{2}$$

$$\sin x \cos y = \frac{\sin(x + y) + \sin(x - y)}{2}$$

2.3. Laws

Law of Sines

$$2r = \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\text{Area} = \frac{1}{2}ab \sin A$$

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

3. Graphs and Geometry

3.1. Lines and Points

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Perpendicular Bisector

$$2(x_2 - x_1)x + 2(y_2 - y_1)y = x_2^2 + y_2^2 - x_1^2 - y_1^2$$

Pythagorean Theorem

$$a^2 + b^2 = c^2 \implies c = \sqrt{a^2 + b^2}$$

3.2. Shapes

Circles

$$(x - m)^2 + (y - n)^2 = r^2$$

$$A = \pi r^2$$

$$C = 2\pi r$$

Regular Polygons

$$\text{Sum of Interior Angles} = 180(n - 2)^\circ$$

$$\text{Area} = \frac{1}{2} \cdot \text{Apothem} \cdot \text{Perimeter}$$

Volumes

$$V_{\text{Cylinder}} = \pi r^2 h$$

$$V_{\text{Cone}} = \frac{1}{3} \pi r^2 h$$

$$V_{\text{Sphere}} = \frac{4}{3} \pi r^3$$

Triangle Similarity

Two triangles $\triangle ABC$ and $\triangle DEF$ are similar if

SSS	SAS	AA
$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$	$\frac{AB}{DE} = \frac{AC}{DF}$ and $\angle A = \angle D$	$\angle A = \angle D$ and $\angle B = \angle E$

3.3. Graph Manipulation

Transformation	Conditions	Description
$g(x) = f(x \pm c)$	(+)	shift left c units
	(-)	shift right c units
$g(x) = f(x) \pm c$	(+)	shift left c units
	(-)	shift right c units
$g(x) = cf(x)$	$c < 0$	vertical compress by a factor of c
	$c > 0$	vertical stretch by a factor of c
$g(x) = f(cx)$	$c < 0$	horizontal stretch by a factor of c
	$c > 0$	horizontal compress by a factor of c
$g(x) = -f(x)$		reflect about the x-axis
$g(x) = f(-x)$		reflect about the y-axis

4. Miscellaneous

4.1. Inverse Functions

Let f and g be two function such that

$$f(g(x)) = x \text{ for all } x \text{ in the domain of } g \quad g(f(x)) = x \text{ for all } x \text{ in the domain of } f$$

Under these conditions, the function g is the *inverse function* of the function f (and vice-versa). The function g can then be denoted as f^{-1} . So,

$$f(f^{-1}(x)) = x \quad f^{-1}(f(x)) = x$$

The domain of f must be equal to the range of f^{-1} , and the range of f must be equal to the domain of f^{-1} .

4.2. Complex Numbers

Let a and b be real numbers. The number $a + bi$ is a *complex number* written in *standard form*. The number a is the *real part* and the number bi is the *imaginary part* of the complex number.

When $b = 0$, the number $a + bi$ is a real number. When $b \neq 0$, the number $a + bi$ is an *imaginary number*. A number of the form bi , where $b \neq 0$, is a *pure imaginary number*.

$$i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, i^n = i^{n \bmod 4}$$

4.3. Relationships

Exponential

$$\begin{array}{ll} a^0 = 1 & \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \\ a^m \cdot a^n = a^{m+n} & (a^m)^n = a^{m \cdot n} \\ \frac{a^m}{a^n} = a^{m-n} & a^{\frac{1}{n}} = \sqrt[n]{a} \\ (ab)^n = a^n \cdot b^n & a^m = a^n \implies m = n \end{array}$$

Logarithmic

$$\begin{array}{ll} \log_a 1 = 0 & \log ab^n = n \log ab \\ \log_a a = 1 & \log a^n b = \frac{1}{n} \log ab \\ \log x = \log_{10} x & \log_n ab = \log_n a + \log_n b \\ \ln x = \log_e x & \log_n \frac{a}{b} = \log_n a - \log_n b \\ y = \log_b x \implies b^y = x & \log_n a = \log_n b \implies a = b \\ a^{\log_a b} = b & \end{array}$$